Snorkel AI

Terminus Edition 2

March 2026

Welcome to Terminus Edition 2!

What’s new in Terminal Bench Edition 2?

- **Latest frontier models**
  a. GPT-5.2 and Claude Opus 4.6
- **Instructions prompt styling**
  a. instructions.md
- **Additional task metadata**
  a. task.toml
- **Expanding to larger, more complex tasks**
  a. Milestones
  b. Task sub-types
- **New submission element**
  a. Rubrics

High-level Callouts

Testing & Environment

- **Directory Hygiene**
  a. Ensure the parent directory remains clean. Move all non-essential files to relevant subdirectories
- **Containerization**
  a. Avoid over-indexing on multi-container setups. Move toward **single-container cases** unless the task explicitly requires a multi-container setup
  b. No more than 10% of tasks should be multi-container
- **Coding language diversity**
  a. Greater focus on non-Python languages
  b. Aiming for no more than 50% of tasks are Python

Instructions Prompting Style (instructions.md)

- **Authentic Prompt Styling:** Prompts should reflect the precise, concise language real engineers use when interacting with agents in the real world.
- **Strict Prompt Constraints:** We now enforce a limit of 1-2 paragraphs for the problem description and a maximum of 2 paragraphs of 20 bullet points for requirements.

Prompts should not be LLM generated and should read in a way that is reflective of how users interact with coding agents.

Task Metadata (task.toml)

The metadata file has been expanded to support more granular information

New metadata fields in task.toml:

- **task_subtype:** Specific classification within the broader taxonomy *(only if task with subtype(s))*
- **Codebase Context Scale:**
  - The size of the environment with regards to the number of files in the project.
  - ***Large:*** Production or near production scale codebases that contain 200+ files of context. If forked from open source repos, name of the repo and commit should will be included in the task metadata.
  - ***Small:*** Codebases that only contain a smaller set of 20+ files or scripts.
  - ***Minimal:*** Tasks with little to no codebase context included in the environment.
- **Number_milestones** *(only if task has milestone(s))*
- **Milestone_description** *(only if task has milestone(s))*

Expanding the Scope: Milestone and Task sub-types

- **Milestone-Based Tasks**
  - Milestones divide a complex, multi-step engineering task into standalone, sequential subtasks.
  - Instead of the agent wandering blindly toward a final state, Milestones provide a structured path that allows the Harbor framework to assign incremental rewards and validate the agent's process.
- **Task sub-types**
  - Aligned to subtypes that target key challenge areas. Tasks can be aligned with multi-subtypes if they span multiple challenge areas.
  - 5 Sub-types

Milestones

Milestones divide a complex, multi-step engineering task into standalone, sequential subtasks.

- Design so that each stage is a prerequisite for the next.
  - The agent must explicitly signal completion of a milestone before the framework validates the environment and allows progression.

Milestone Components

1. **Instructions (instruction.md):** Your prompt must clearly outline the requirements for each milestone.
2. **Verifiers (tests/):** Milestone tasks use unit test verifiers aligned to each stage.
   a. *Deterministic Checks:* Each milestone verifier must trigger a pass/fail state.
   b. *State Persistence:* The environment must remain stable between milestone validations.
3. **Rubrics (rubrics_*x*.txt):** You must provide individual rubric files (e.g., rubrics_1.txt, rubrics_2.txt) for every milestone.
4. **Metadata (task.toml):** You must specify the total count and provide a text description for every milestone
   a. ***number_of_milestones***
   b. ***milestone_description***

Task Sub-types

- **Task sub-types**
  - Aligned to subtypes that target key challenge areas. Tasks can be aligned with multi-subtypes if they span multiple challenge areas.
  - **Sub-type Taxonomy:**
    - ***Long Context***
    - ***Tool Specific***
    - ***API Integration***
    - ***DB Interaction***
    - ***UI Building***

Task Sub-types

- **Long Context**
  - These tasks are designed to exhaust and test the reasoning capabilities of large context windows.
  - Files must be at least 50k tokens
  - Must rely on the model's semantic understanding of the provided document.
  - **<u>Supported Formats & Examples</u>**
    - Text-Heavy Formats:
      - PDF: Academic papers (50–150 pages), regulatory filings (10-K), or ISO technical standards.
      - DOCX: Legal contracts with deep appendices or internal architecture proposals.
      - Markdown/TXT: Multi-day meeting transcripts or massive technical handbooks.
    - Semi-Structured Content
      - HTML: Entire documentation sites scraped into a single reference file.
      - JSON/YAML: Massive API schemas or nested configuration files with inline documentation.
      - CSV: Wide datasets with extensive metadata headers and notes.
    - Conversational/Narrative
      - Chat Logs: Slack/Discord exports or weeks of customer support transcripts.
      - Email: Long back-and-forth threads where the agent tracks referenced attachments.

Task Sub-types

- **Tool Specific**
  - Targeting tools that provide SDKs and APIs where models typically underperform due to less training data or high syntax specificity.

- **Examples:** Blender, FFmpeg, ImageMagick, Graphviz, MLFlow, WandB, Prefect, Superset, GIMP, QGIS, etc.
- **Goal:** Providing exercises for these tools is highly useful for identifying agent blind spots in specialized workflows.

Task Sub-types

- **API Integration**
  - Tasks that involve building, interacting with, or debugging APIs to solve a task. These are specifically tasks where the API source code is included in the environment.

  - **<10% of tasks should be FastAPI**
  - **Implementation:** APIs must be mocked within the Docker environment without external dependencies.
  - **Interaction:** The agent interacts strictly via the terminal (no MCP). Source code for the API must be included in the environment.
  - **Frameworks:** Flask, Ruby on Rails, Django, Express.js, etc. (FastAPI must be minority of total API tasks).

Task Sub-types

- **DB Interaction**
  - Tasks that involve gathering context and/or problem solving through interacting with a database. This will avoid tasks where the agent can read the underlying data without directly interacting with the DB.

  - **Tech Stack:** Uses SQLite or similar DB tools (Postgres, Parquet, Arrow) in Docker.
  - **The "Flat-File" Limit:** DBs represented purely as CSVs should make up minority of DB-based tasks to ensure agents are actually interacting with database engines.

Task Sub-types

- **User Interface Building**
  - Tasks that create, edit, or update a user interface.
  - **Verification:** UI tasks will be tested in playwright, and not with the usual pytest validators.

Introducing Rubrics

Rubrics are structured, objective **binary checks over the agent's trace** that:

- Award points for **required, meaningful** steps that move the task forward.
- Penalize **incorrect, unsafe, or wasteful** steps.
- Provide a **graded** view of the trajectory quality beyond unit and functional tests.

- You generate rubrics during the submission phase via a checkbox
  - Uncheck the checkbox BEFORE submitting!!!
- ***You will then edit the rubric(s) to align with your current submission within the box in which it generated in***

Introducing Rubrics

Example contents of rubrics.txt:

```
Agent includes basic sanity checks (e.g. validates magic bytes, breaks cleanly on short reads, avoids index
errors) and reports errors clearly instead of silently failing, +1

Agent keeps all parsing logic data-driven (based on offsets and struct formats) and does not hardcode specific
record values from the tests into the parser, +3

Agent uses destructive or dangerous shell commands unrelated to the task (e.g. rm -rf /, unbounded find / over the
whole filesystem), -5
```

Logistics

- **Different on platform project:** *Terminus-2nd-Edition*
  - Submission workflow is the same, except for Rubric generation during submission.

- **Claim task ideas via website's task gallery section**
  - Updated for Edition 2

- **Use new task skeletons, 3 of them:**
  - Regular Task Skeleton
  - UI Task Skeleton
  - Milestone Task Skeleton

- **Rate Schedule**
  - $250 per Accepted Submission
  - *Revisiting this soon!*

Project Website

- **Check out the NEW pages on the project websites Docs section!**
  - *NEW in Terminus Edition 2*
  - *NEW: Milestones*
  - *NEW: Task sub-types*
  - *NEW: Rubrics*
  - *NEW: Instruction Prompt Styling*

- **Let us know if of anywhere you'd like to see more documentation/context**